

Data Dictionary & Cheat Sheet

Project 3 — Merchandising & Product Profitability | Contoso Retail Group

Every table and every column in **your** data file, explained.

The business

Contoso Retail Group is a consumer-electronics retailer trading in **eight countries** (United States, Canada, United Kingdom, Germany, France, Italy, Netherlands, Australia) through **73 physical stores**, an online shop, and a Click & Collect service. The data covers **1 January 2015 to 31 March 2024**. The extract was taken on **30 April 2024**.

Rules that apply to this data

Rule	What it means for you
1. Money is in local currency	Every amount in sales and returns is in the currency the store trades in — a German store records euros, an Australian store records Australian dollars. To compare or add amounts across countries you must first convert to a single currency: USD amount = local amount ÷ ExchangeRate . The prices in product are already in USD.
2. Two fact tables, no join between them	sales and returns are separate tables at different grains, and there is no relationship between them . They meet only through the dimensions they share: Date, store and product. A measure that mixes the two — return rate, for instance — works because both sides are filtered by the same dimension, not because the tables are joined.
3. A return is not a cancelled sale	A returned line still exists in sales — the goods were bought and paid for. The returns table records what came back afterwards, sometimes only part of the line. So never subtract returns from sales rows ; report revenue and returned value separately and let the reader see both. 9,919 lines were returned out of 121,160 sold.

A note on the discount column. sales[Discount] is a *rate*, not an amount: 0.15 means the line was sold at 15% off. The money you gave away is (UnitPrice – NetPrice) × Quantity. Watch the scale carefully when you clean this column — a value of 15 instead of 0.15 will make your discount measure a hundred times too large, and the error is not obvious on a chart.

sales **FACT**

One row per **product line** on an order. An order with three different products has three rows, all sharing the same OrderKey. About 121,000 rows.

Column	Data type	What it means	Notes
OrderKey	Whole number	The order the line belongs to. Repeats across the lines of the same order.	Count orders with DISTINCTCOUNT, never COUNTROWS.
LineNumber	Whole number	Position of the line inside its order, starting at 1.	OrderKey + LineNumber together are unique.

Column	Data type	What it means	Notes
OrderDate	Date	The day the customer placed the order.	The main date for time analysis.
StoreKey	Whole number	Which store the sale belongs to. Online and Click & Collect orders are still credited to a store.	Links to store.
ProductKey	Whole number	Which product was sold.	Links to product.
Quantity	Whole number	How many units of that product were bought on this line.	1 to 10.
UnitPrice	Decimal	Full list price of one unit, in the store's local currency , before any discount.	= product[Price] × ExchangeRate.
NetPrice	Decimal	Price actually charged per unit after the discount, local currency .	= UnitPrice × (1 – Discount). Use this for revenue.
UnitCost	Decimal	What one unit cost Contoso to buy, local currency .	Use this for cost and margin.
CurrencyCode	Text	The currency this row is recorded in.	USD, EUR, GBP, CAD or AUD.
ExchangeRate	Decimal	Units of local currency per 1 US dollar.	Divide by this to get USD. Always 1 for USD.
Discount	Decimal	Discount rate applied to the line, as a fraction.	0.15 means 15% off. 0 means no discount.
Channel	Text	How the order was placed: In-Store, Online or Click & Collect.	Constant across all lines of one order.
PromotionKey	Whole number	Which campaign gave the discount.	0 means no promotion — it is a real row, not a blank.

returns **FACT**

One row per **returned sales line**. 9,919 rows. It carries OrderKey and LineNumber so you can trace a return back to its sale, but you do **not** need a relationship on them — see rule 3.

Column	Data type	What it means	Notes
ReturnKey	Whole number	A unique identifier for this return record.	Every row has its own. Not used in any relationship.
OrderKey	Whole number	The order the line belongs to. Repeats across the lines of the same order.	Count orders with DISTINCTCOUNT, never COUNTROWS.
LineNumber	Whole number	Position of the line inside its order, starting at 1.	OrderKey + LineNumber together are unique.
StoreKey	Whole number	Which store the sale belongs to. Online and Click & Collect orders are still credited to a store.	Links to store.
ProductKey	Whole number	Which product was sold.	Links to product.

Column	Data type	What it means	Notes
ReturnDate	Date	When the return was received.	Blank unless IsReturned is TRUE.
ReturnedQuantity	Whole number	How many of the units were sent back.	Can be less than Quantity (a partial return).
NetPrice	Decimal	Price actually charged per unit after the discount, local currency.	= UnitPrice × (1 – Discount). Use this for revenue.
CurrencyCode	Text	The currency this row is recorded in.	USD, EUR, GBP, CAD or AUD.
ExchangeRate	Decimal	Units of local currency per 1 US dollar.	Divide by this to get USD. Always 1 for USD.
ReturnReason	Text	Why it came back: Changed mind, Faulty, Damaged, Wrong item, Late delivery.	"Late delivery" only ever appears on Online orders.

product **DIMENSION**

One row per product. 2,517 products. The brand is **not** here — product holds only ManufacturerKey, and the brand details live in their own table.

Column	Data type	What it means	Notes
ProductKey	Whole number	Which product was sold.	Links to product.
ProductName	Text	Full product name.	Unique for every product.
ManufacturerKey	Whole number	Unique brand identifier.	Links product to manufacturer.
CategoryName	Text	The product category, e.g. Computers, Home Appliances.	8 categories.
SubCategoryName	Text	The narrower product group, e.g. Desktops, Televisions.	32 subcategories.
Color	Text	Product colour.	16 colours.
WeightUnit	Text	The unit Weight is expressed in: grams, kilograms or ounces.	Read this before using Weight — see rule 2.
Weight	Decimal	Product weight in the unit named by WeightUnit.	Not additive. Convert first.
Cost	Decimal	What Contoso pays the supplier per unit, in USD .	The USD basis of sales[UnitCost].
Price	Decimal	Recommended retail price per unit, in USD .	The USD basis of sales[UnitPrice].

manufacturer DIMENSION

One row per brand. 11 rows. Two of its columns exist nowhere else and are the point of the table: BrandTier and IsOwnBrand let you compare Contoso's own label against the brands it resells.

Column	Data type	What it means	Notes
ManufacturerKey	Whole number	Unique brand identifier.	Links product to manufacturer.
ManufacturerName	Text	The brand name, e.g. Contoso, Ltd.	11 brands, all fictional.
SupplierCountry	Text	Where the brand is supplied from.	Seven countries.
BrandTier	Text	Premium, Mainstream, Value, or Own Brand.	Assigned from the brand's own average shelf price. Own Brand is Contoso's own label.
IsOwnBrand	True/False	TRUE only for Contoso, Ltd — the house label.	The margin and returns comparison of own-brand against resold brands is a core question in this project.

store DIMENSION

One row per store. 73 stores in total.

Column	Data type	What it means	Notes
StoreKey	Whole number	Which store the sale belongs to. Online and Click & Collect orders are still credited to a store.	Links to store.
GeoAreaKey	Whole number	Which region the row belongs to.	Links to geography.
CurrencyCode	Text	The currency this row is recorded in.	USD, EUR, GBP, CAD or AUD.
OpenDate	Date	The day the store opened for business.	No sale exists before this date.
CloseDate	Date	The day the store closed, if it has closed.	Blank for stores still trading.
SquareMeters	Whole number	Selling floor area in square metres.	245 to 3,500. Use for revenue per m ² .
Status	Text	Open, Closed, or Restructured.	"Restructured" means the store closed and reopened under a new StoreKey.

geography DIMENSION

One row per state, province or region. 595 rows. Only store points at it in this version, so a country slicer built from it filters by **store** country.

Column	Data type	What it means	Notes
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Column	Data type	What it means	Notes
GeoAreaKey	Whole number	Which region the row belongs to.	Links to geography.
Continent	Text	Europe, North America or Oceania.	
CountryCode	Text	Two-letter country code.	US, CA, GB, DE, FR, IT, NL, AU.
CountryFull	Text	Full country name.	Use this on maps, not CountryCode.
State	Text	State, province, county or region.	Granularity differs by country: US states, UK counties, Italian provinces, French regions.

promotion **DIMENSION**

One row per campaign, plus one row for "no promotion". 65 rows.

Column	Data type	What it means	Notes
PromotionKey	Whole number	Which campaign gave the discount.	0 means no promotion — it is a real row, not a blank.
PromotionName	Text	Campaign name, e.g. "Black Friday 2022".	Seven campaigns repeat every year.
PromotionType	Text	Clearance, Seasonal, Flash, or No Promotion.	Flash = Black Friday and Cyber Week.
CategoryScope	Text	Which category the campaign applied to, or "All".	e.g. Back to School only discounts Computers.
Discount	Decimal	Discount rate applied to the line, as a fraction.	0.15 means 15% off. 0 means no discount.
StartDate	Date	First day the campaign ran.	Blank on the No Promotion row.
EndDate	Date	Last day the campaign ran.	Blank on the No Promotion row.

currency DIMENSION

One row per trading currency. 5 rows. Joins to store, not to either fact.

Column	Data type	What it means	Notes
CurrencyCode	Text	The currency this row is recorded in.	USD, EUR, GBP, CAD or AUD.
CurrencyName	Text	Full currency name, e.g. Pound Sterling.	
Symbol	Text	Currency symbol.	For display in report titles.
PrimaryMarket	Text	The main country using it.	

What is deliberately not in the file

You are expected to build these yourself — they are part of the assessment:

- **A Date table.** Nothing in the workbook is a calendar. Build one in DAX with `CALENDAR`, add the columns you need.
- **Delivery lag.** `DeliveryDate - OrderDate`.
- **Customer tenure.** `From SignupDate`.